

Cardiovascular Disease and Metabolic Abnormalities in the Kidney Transplant Recipient

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CARDIOVASCULAR DISEASE

Cardiovascular disease (CVD) is the most frequent cause of death with a functioning kidney allograft. The US Renal Data System (USRDS) 2014 annual data report revealed that mortality from CVD among kidney transplant recipients (kidney alone or combined organ transplant) is nearly twice that observed for infection or malignancy. Cardiovascular (CV) risk factors in kidney transplant recipients encompass both conventional and unconventional risk factors. Although transplantation ameliorates some CV risks by restoring kidney function, it introduces new CV risks, including impaired glucose tolerance or diabetes mellitus (DM), hypertension, and dyslipidemia, which are derived in part from immunosuppressive medications. [Box 111.1](#) summarizes suggested CV risk factors in transplant recipients. Selected risk factors are also discussed.

CONVENTIONAL CARDIOVASCULAR DISEASE RISK FACTORS

Posttransplantation Hypertension

Hypertension is present in more than 50% of kidney transplant recipients, and the true proportion may be much higher. In one single-center study, ambulatory blood pressure (BP) measurements demonstrated that only 5% of kidney transplant recipients were normotensive (defined as BP < 130/80 mm Hg).¹ Another study suggests that clinic BP measurements (vs. ambulatory BP monitoring) may underestimate the prevalence of hypertension in kidney transplant recipients; of 76 patients studied, 22.4% had masked hypertension (defined as BP > 130/80 mm Hg).²

Hypertension is a risk factor for both CVD and kidney graft failure. A retrospective cohort study revealed a graded risk for graft failure and death with increasing levels of systolic BP (each 10 mm Hg of systolic BP increase was associated with an increased relative risk for graft failure [risk ratio (RR), 1.12; $P < .000$], death-censored graft failure [RR, 1.17; $P < .0001$], and death [RR, 1.18; $P < .0001$]).³ Post hoc analysis of the Folic Acid for Vascular Outcome Reduction in Transplantation (FAVORIT) study showed that each 20 mm Hg increase in baseline systolic BP was associated with a 32% increase in CVD risk and a 13% increase in mortality risk. However, for every 10 mm Hg decrease in diastolic BP below 70 mm Hg, a 31% increase in CVD and mortality risk was observed. For diastolic BP levels greater than 70 mm Hg, no significant association between diastolic BP and outcomes was observed.⁴

Risk factors for posttransplantation hypertension include preexisting hypertension, cyclosporine (and, to a lesser degree, tacrolimus), corticosteroids, various donor-related factors (e.g., donor age, donor

hypertension, donor family history of hypertension), delayed graft function, chronic allograft injury, high body mass index or excess weight gain, acute rejection episodes, recurrent or de novo glomerulonephritis, and transplant renal artery stenosis. Sodium intake after transplantation and excess renin output from the native kidneys also may contribute in some patients. In kidney transplant recipients with severe hypertension refractory to medical therapy, bilateral native nephrectomy may ameliorate BP control. In a retrospective study of kidney transplant recipients with autosomal dominant polycystic kidney disease, ipsilateral native nephrectomy at the time of allograft transplant was associated with a significant decrease in the use of antihypertensive drugs needed for hypertension control compared with kidney transplantation alone at 12-, 24-, and 36-month follow-up. Subsequent contralateral native nephrectomy further improved BP control.⁵

The American College of Cardiology (ACC)/American Heart Association (AHA) guidelines on BP management advocate liberalizing the BP target to less than 160/90 mm Hg in the early posttransplantation period to maintain organ perfusion and avoid hypotension to reduce the risk of graft thrombosis. Beyond the first posttransplantation month, more stringent BP control to less than 130/80 is recommended to prevent end-organ damage.⁶ The Kidney Disease: Improving Global Outcome (KDIGO) guidelines suggest a BP goal of less than 130/80 mm Hg for kidney transplant recipients irrespective of the level of albuminuria. However, such recommendations are based solely on epidemiologic data because no randomized trials demonstrate the optimal BP target in kidney transplant recipients. A BP goal of less than 125/75 mm Hg for patients with proteinuria is of uncertain benefit. Management of posttransplantation hypertension should include identification and treatment of the underlying cause, lifestyle modifications (see [Chapter 36](#)), and treatment of associated CV risk factors.

We recommend perioperative β -blockers in transplant candidates with a history of coronary artery disease because they reduce coronary artery disease (CAD) events. Diuretics are frequently used in the early posttransplantation period in patients who are volume expanded. Most transplant centers use calcium channel blockers for initial therapy because of their added benefit of antagonizing calcineurin inhibitor (CNI)-induced afferent arteriolar vasoconstriction, and their demonstrated efficacy in the general population irrespective of age, sex, and salt intake.⁷ The nondihydropyridine calcium channel blockers diltiazem and verapamil are cytochrome CYP3A4 inhibitors, and their use also permits CNI or mammalian target of rapamycin (mTOR) inhibitor dose reductions. Angiotensin-converting enzyme (ACE) inhibitors and angiotensin receptor blockers (ARBs) can cause acute changes in renal function and hyperkalemia and hence are usually not started until allograft function is stable. Transplant renal artery

BOX 111.1 Cardiovascular Risk Factors in Kidney Transplant Recipients

Conventional Modifiable

Hypertension
Dyslipidemia
Obesity or metabolic syndrome
Smoking

Nonmodifiable

Age
Family history
Male sex
History of pretransplant and posttransplant cardiovascular disease
Race/ethnicity (White ethnicity)

Unconventional Modifiable or Potentially Modifiable

Proteinuria
Posttransplantation diabetes mellitus
Left ventricular hypertrophy
Inflammatory markers (e.g., inflammatory cytokines, C-reactive protein)
Anemia
Hyperuricemia
Cytomegalovirus infection
Delayed graft function
Hyperhomocysteinemia
Time on dialysis
Chronic kidney disease (posttransplant)
Low albumin
Obstructive sleep apnea
Prothrombotic state
Low physical activity

Nonmodifiable

Recipient Factors

Prior acute rejection episodes
Preexisting coronary calcification
Cardiac troponin T

Donor Factors (Donor Quality)

Donor age
Donor hypertension
Donor recipient size mismatch
Donor *APOL1* gene polymorphism (may be associated with early graft dysfunction)

stenosis should be considered when serum creatinine rises to over 30% above baseline (see Chapter 43). Limited studies in kidney transplant recipients suggest that renin-angiotensin-aldosterone system (RAAS) blockade reduces CV events and ameliorates CNI-induced interstitial fibrosis and tubular atrophy. However, the beneficial effect of ACE inhibitors or ARBs on patient or graft survival has not been consistently demonstrated.⁸ A meta-analysis of eight randomized trials demonstrated that RAAS inhibitors did not significantly alter all-cause mortality (RR, 0.96; 95% confidence interval [CI], 0.62–1.51), graft failure (RR, 0.76; 95% CI, 0.49–1.18), or doubling of serum creatinine (RR, 0.84; 95% CI, 0.51–1.39) compared with placebo (with similar BP control) or standard of care.⁹ Because of the lack of conclusive evidence that one class of antihypertensive agent is superior to another in transplant recipients, treatment should be individualized based on efficacy, tolerability, concomitant comorbidity, and drug-drug interactions with immunosuppressive agents. Potential advantages and

disadvantages of different classes of antihypertensive agents in kidney transplant recipients are shown in Table 111.1.

Posttransplantation Dyslipidemia

Dyslipidemia is common after transplantation, in part because of the hyperlipemic effect of corticosteroids, cyclosporine, tacrolimus, and the mTOR inhibitors sirolimus and everolimus. Sirolimus and everolimus are associated with the worst lipid profiles, followed by cyclosporine, and to a lesser extent tacrolimus. One single-center study demonstrated that the mean values of total cholesterol, low-density lipoprotein (LDL) cholesterol, and triglycerides (TGs) and the incidence of CAD were higher among patients receiving mTOR inhibitors compared with those receiving CNI-based immunosuppression (controls). However, the risk for CV events is not significantly higher among patients receiving mTOR inhibitors compared with controls.^{10,11} A substudy of the MECANO trial also found no difference in CV outcomes among kidney transplant recipients randomized to early cyclosporine (CSA) to everolimus conversion therapy ($n = 96$) compared with those remaining on CSA-based immunosuppression ($n = 89$) over a 7-year follow-up period.¹² Studies in heart transplant recipients suggest that CNI to sirolimus conversion therapy may attenuate cardiac allograft vasculopathy (CAV) progression and CAV-related events compared with CNI continuation independent of LDL levels.¹³ The antiproliferative and cardioprotective effects of mTOR inhibitors and the reduction in CNI-related risk factors may offset the adverse effects of mTOR inhibitor-associated hyperlipidemia.

Other etiologic factors for posttransplant dyslipidemia include age, diet, rapid weight gain, hyperinsulinemia, preexisting hypercholesterolemia, allograft dysfunction, proteinuria, genetic predisposition, and the use of diuretics and β -blockers (nonselective > selective β_1 -blockers).

Management of Hyperlipidemia

Lifestyle changes (diet and exercise) and statins remain the cornerstone of therapy. In addition to their lipid-lowering effect, statins may protect against CVD via their antiproliferative and anti-inflammatory properties and ability to reduce circulating endothelin-1, C-reactive protein levels, systolic and diastolic BP, and pulse pressure. The benefits of statins in the general population have been demonstrated in several large randomized controlled trials (RCTs). The ALERT trial, the only prospective RCT in transplant recipients comparing statins (fluvastatin) with placebo, found a beneficial effect of early initiation of fluvastatin on posttransplant CV outcome. Patients who received statin therapy within the first 4 years after transplantation had a risk reduction of 64% compared with 19% for those who received therapy after 10 years.¹⁴

Box 111.2 summarizes different guidelines for the management of posttransplantation dyslipidemia.^{15,16} The different classes of lipid lowering-agents are discussed in the following sections.

Statin Therapy

Kidney transplantation is considered a risk factor for CAD similar to other conventional cardiac risk factors. Despite the lack of evidence-based recommendations, most transplant centers treat dyslipidemias aggressively with statin therapy along with lifestyle modification. The concomitant use of statins and cyclosporine often results in a several-fold increase in statin blood level because CSA inhibits CYP450 3A4 enzyme, of which statins are major substrates. In addition, CSA inhibits OATP1B1/SLCO1B1-mediated hepatic uptake of statins, resulting in significant medication interaction. Therefore, in CSA-treated patients, all statins should be introduced at low doses and uptitrated

TABLE 111.1 Potential Advantages and Disadvantages of Different Classes of Antihypertensive Agents

Classes of Drugs	Advantages/Beneficial Effects	Disadvantages/Adverse Effects
ACE inhibitors, ARBs	- Beneficial in patients with proteinuria,^a DM, LVH - Potential renoprotective and cardioprotective effects - Beneficial in posttransplantation erythrocytosis - Losartan may lower serum uric acid levels^b	- Potential worsening anemia - GFR decline, hyperkalemia
Mineralocorticoid receptor antagonists (e.g., spironolactone, eplerenone)	- Improve outcome in patients with HFrEF - May improve blood pressure control in difficult-to-treat hypertension	Severe hyperkalemia when used in combination with ACE inhibitor or ARB, particularly in patients with poor kidney function
α-Blockers (e.g., prazosin)	Beneficial in patients with benign prostatic hypertrophy or neurogenic bladder	Orthostatic hypotension
β-Blockers (e.g., carvedilol, metoprolol, bisoprolol)	Beneficial in patients with CAD or ischemic heart disease	Blunting of hyperglycemic unawareness, erectile dysfunction, hyperlipidemia, hyperkalemia (nonselective > selective β1-blockers), bronchospasm
Calcium channel blockers: nondihydropyridine (diltiazem, verapamil) and dihydropyridine (amlodipine, nifedipine)	- Ameliorate CNI-induced vasoconstriction - Diltiazem may permit CNI dose reduction by up to 40% - Verapamil by 30%–50%	
Central α-agonists (e.g., clonidine, methylodopa)	Beneficial in patients with diabetic autonomic dysfunction	
Diuretics (loop and thiazide)	- Beneficial in patients who are volume expanded - Thiazide may improve hyperkalemia commonly seen in CNI-treated patients. Its use may also increase BMD decrease fracture risk (beneficial in osteoporosis).	
Direct vasodilators (e.g., hydralazine, minoxidil)	- Minoxidil: may be beneficial in difficult-to-treat hypertension - Hydralazine/nitrate combination therapy is recommended as add-on therapy in African Americans with refractory HFrEF and NYHA class III-IV heart failure despite standard of care therapy	

^aThe 2009 KDIGO clinical practice guidelines recommend an ACE inhibitor or ARB as first-line therapy for patients with hypertension and proteinuria of 1 g/day or greater.

^bMay be beneficial in patients with gout.

ACE, Angiotensin-converting enzyme; ARBs, angiotensin receptor blockers; CAD, Coronary artery disease; CNI, calcineurin inhibitor; DM, diabetes mellitus; GFR, glomerular filtration rate; HFrEF, heart failure with reduced ejection fraction; LVH, left ventricular hypertrophy; NYHA, New York Heart Association.

at the discretion of the clinicians. Maximal doses should be avoided because CSA-induced increase in statin bioavailability can result in an increased risk for myopathy and rhabdomyolysis. In contrast, tacrolimus does not inhibit CYP3A4 to a significant extent and has no effect on OATP1B1/SLCO1B1 and thus drug-drug interaction with statins is not observed. However, rhabdomyolysis associated with tacrolimus and statin use may be seen in patients on concomitant diltiazem therapy because the latter is a moderate CYP3A4 inhibitor.

Non-Statins Drug Therapy

Ezetimibe. Ezetimibe and statin combination therapy may improve cholesterol control because of their complementary mechanisms of action. Ezetimibe blocks intestinal absorption of dietary cholesterol, whereas statin inhibits hepatic cholesterol synthesis. Results of the SHARP trial in which CKD patients were randomized to receive simvastatin 20 mg plus ezetimibe 10 mg ($n = 4650$) or placebo ($n = 4620$) demonstrated that simvastatin and ezetimibe combination therapy conferred a 17% risk reduction compared with placebo at a median follow-up of 4.9 years.¹⁷ Limited studies suggest that statin and ezetimibe combination therapy is safe and effective in the treatment of dyslipidemia in kidney transplant patients who are refractory to statin therapy. In a study of 67 patients with posttransplantation hyperlipidemia resistant to statins, treatment with ezetimibe alone or with ezetimibe and statin significantly reduced total cholesterol and LDL cholesterol by 25% and 34%, respectively, during the first month of treatment.¹⁸ The 2019 European Society of Cardiology (ESC)/European Atherosclerosis Society (EAS) guidelines advocate adding

ezetimibe to statin therapy for solid organ transplant recipients who are refractory to the highest tolerated statin dose or as a second-line agent in those who are intolerant to statin. The AHA/ACC guidelines recommend adding ezetimibe to statin therapy in very high CVD risk patients if LDL-C level remains at least 70 mg/dL despite maximally tolerated statin therapy.

Protein convertase subtilisin/kexin type 9 inhibitors (PCSK9).

PCSK9 is a proprotein convertase involved in the degradation of LDL receptors in the liver. PCSK9 inhibitors are human monoclonal antibodies that inhibit the binding of PCSK9 to the LDL receptors. Two large randomized, placebo-controlled trials involving high CV risk patients demonstrated that the PCSK9 inhibitors evolocumab (FOURIER trial, $n = 27,564$) and alirocumab (ODYSSEY trial, $n = 18,924$) reduced LDL-C levels significantly more than statins and significantly decreased CV morbidity and mortality (CV death, nonfatal myocardial infarction [MI], or stroke) independent of baseline LDL-C levels. Post hoc analysis of the ODYSSEY and FOURIER trials demonstrated that the efficacy and safety of alirocumab and evolocumab were comparable among subjects with and those without estimated glomerular filtration rate (eGFR) less than 60 mL/min/1.73 m².¹⁹ The role of PCSK9 inhibitors in the treatment of dyslipidemia or CV risk reduction in chronic kidney disease (CKD) patients and in kidney transplant patients remains to be studied.

Management of Hypertriglyceridemia

Initial management should include lifestyle changes, increased physical activity, weight reduction, dietary counseling (carbohydrate restriction),

BOX 111.2 Pharmacologic Therapy for Dyslipidemia in Kidney Transplant Recipients

KDIGO Guidelines

- Statins in all adult transplant recipients.
- In the absence of traditional risk factors, transplant recipients <30 years old could choose not to receive statins if priority is placed on minimizing polypharmacy and drug toxicity over the small absolute risk reduction.^a
- Elevated TGs:
 - Therapeutic lifestyle changes: low-fat diet (<15% total calories), reduction of mono- and disaccharide intake, reduction in total amount of dietary carbohydrates, use of fish oil to replace some long-chain TGs
 - Fibrate use to prevent pancreatitis or reduce CV risk not recommended

ACC/AHA Guidelines

- Statin therapy should be based on clinical judgment (weigh risk and benefits) and patient preference
- Follow-up lipid profile at 4 and 12 weeks after treatment assess medical adherence, then every 3–12 months as clinically indicated
- Benefits outweigh risk in patients with:
 - Clinical ASCVD
 - LDL-C > 190 mg/dL
 - DM, age 40–75 years, LDL-C 70–189 mg/dL, no clinical ASCVD
 - Clinical ASCVD or diabetes, LDL-C 70–189 mg/dL, and estimated 10-year ASCVD risk > 7.5%
- Elevated TGs not addressed; however, in nontransplant setting, ACC/AHA guidelines suggest adding n-3 polyunsaturated fatty acid (icosapent ethyl) to statin therapy in high-risk patients with TG ≥ 150 mg/dL^b

2019 ESC/EAS Guidelines

- Treatment of dyslipidemias in transplant recipients similar to that for patients at high or very high CV risk
- Statins should be considered first-line agents
- Consider ezetimibe or ezetimibe-statin dual therapy in patients who are intolerant of statins or those with significant dyslipidemia despite maximally tolerated statin treatment, respectively^c
- Elevated TGs^d: consider adding omega-3 fatty acid to statin therapy in high-risk patients with TG 135–499 mg/dL (currently insufficient evidence to recommend routine use of fish oil in the transplant population)

^aRationale for recommendations: risk of coronary events is age dependent, and the ALERT trial did not enroll subjects <30 years.

^bApproved by the US Food and Drug Administration for use as an adjunctive therapy to maximally tolerated statins to reduce the risk of CV events in adults with TG > 150 mg/dL. Patients must also have established CV disease or DM, and two more risk factors for CV disease.

^cCyclosporine can induce a 2- to 12-fold increase in the ezetimibe level. Concomitant ezetimibe and cyclosporine therapy can also raise cyclosporine concentration by 15%.

^dSuggested lifestyle changes: reduce excessive body weight; increase physical activity; reduce alcohol, carbohydrate, and mono- and disaccharide intake; replace saturated fats with mono- or polyunsaturated fats. ACC, American College of Cardiology; AHA, American Heart Association; ASCVD, atherosclerotic cardiovascular disease; CV, cardiovascular; DM, diabetes mellitus; EAS, European Atherosclerosis Society; ESC, European Society of Cardiology; KDIGO, Kidney Disease Global Outcomes; LDL, low-density lipoprotein; TGs, triglycerides.

smoking cessation, and correction of secondary causes. The latter may include untreated DM, excess alcohol intake, or drug-induced causes. Severe hypertriglyceridemia (TG level > 500 mg/dL or 5.65 mmol/L) may occur with the use of sirolimus and everolimus. Management includes dose reduction, pharmacologic therapy, and, in refractory cases, switching of mTOR inhibitors to mycophenolic acid derivatives or tacrolimus.

Drug Therapy for Hypertriglyceridemia

Fibrates. Fibrates have the most pronounced effect on lowering plasma TG levels of available lipid-lowering agents. Of available options (bezafibrate, ciprofibrate, fenofibrate, and gemfibrozil), the first three can increase serum creatinine in cyclosporine-treated patients. Although all fibrates have been associated with creatinine kinase elevations (with or without overt rhabdomyolysis and myopathy) when used in combination with statins, gemfibrozil may have a greater risk for myopathy compared with fenofibrate or bezafibrate. In kidney transplant recipients, a daily dose exceeding 900 mg for gemfibrozil, 400 mg for bezafibrate, and 200 mg for fenofibrate is not recommended.¹⁶ Fenofibrate is contraindicated in patients with eGFR less than 30 mL/min/1.73 m². Data on the safety of gemfibrozil in patients with advanced CKD are lacking.

Ezetimibe. Limited studies demonstrated that ezetimibe has TG-lowering effects and is safe and effective in the transplant settings.

Bile acid sequestrants. Bile sequestrants are currently seldomly used given the available wide selection of highly effective lipid-lowering agents with more favorable side effect profiles.¹⁶ Notably, studies in the general population suggest that bile acid sequestrants may increase TG levels. Their use is contraindicated in individuals with TGs greater than 400 to 500 mg/dL.

Niacin. Two large RCTs showed that nicotinic acid use is of no benefit but is associated with serious adverse effects. Currently, no medication containing nicotinic acid is approved for use in Europe.²⁰

Fish oil. The ESC/EAS guidelines recommend adding omega-3 polyunsaturated fatty acids (2 g icosapent ethyl twice daily) to statin therapy in high-risk patients with TG levels between 135 to 499 mg/dL. There is insufficient evidence to recommend routine use of fish oil in the kidney transplant population.¹⁶

Summary

Statins should be the first-line therapy for the treatment of nonhigh-density lipoprotein (non-HDL) cholesterol because of their well-established safety and efficacy in preventing CVD in the general population. Ezetimibe should be used as a second-line agent in patients intolerant of statins or in combination therapy with statin when the LDL-C level remains at least 70 mg/dL despite maximally tolerated statin therapy. Adherence to lifestyle changes and to statin therapy should be reemphasized before adding a nonstatin lipid-lowering agent to statin therapy. In patients with severe hypertriglyceridemia, adding omega-3 polyunsaturated fatty acids or fibrate to statin therapy may be considered. However, caution should be exercised because of potential fibrate and statin drug-drug interactions. The choice of one lipid-lowering agent over the other should be based on adverse effects and potential drug-drug interactions. Simvastatin and gemfibrozil combination therapy is associated with an increased risk for rhabdomyolysis and should be avoided. Whether PCSK9 inhibitors are safe and effective in the transplant setting remains to be studied.

POSTTRANSPLANTATION DIABETES MELLITUS (NEW-ONSET DIABETES AFTER TRANSPLANTATION)

Posttransplantation DM (PTDM) occurs in 4% to 25% of kidney transplant recipients. Variations in incidence may arise from differences in definition, duration of follow-up, and the presence of both modifiable and nonmodifiable risk factors. Over the years, PTDM has undergone changes in nomenclatures including steroid diabetes, PTDM, new-onset DM, transplant-associated hyperglycemia, and new-onset diabetes after transplantation (NODAT).²¹ In 2014 the International Expert Panel recommended changing the terminology

Risk Factors for Posttransplantation Diabetes Mellitus¹

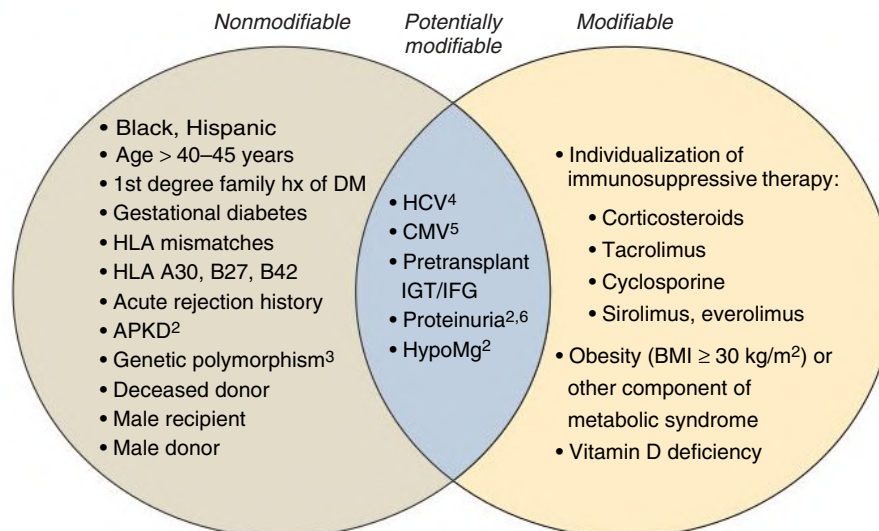


Fig. 111.1 Risk Factors for Posttransplantation Diabetes Mellitus. ¹Restoration of insulin metabolism by a functioning graft may unmask pretransplant impaired glucose tolerance or DM. ²The associations between APKD, proteinuria, and hypomagnesemia and PTDM have not been consistently observed. ³Three genetic variants significantly associated with PTDM in random effects meta-analysis include CDKAL1 rs10946398, KCNQ1 rs2237892, and TCF7L2 rs7903146. ⁴Antiviral therapy with direct-acting antiviral agents is discussed in [Chapter 110](#). ⁵Posttransplant CMV prophylactic and preemptive therapy is discussed in [Chapter 110](#). ⁶One study showed that moderately elevated early posttransplantation albuminuria is a risk factor for PTDM in a dose-dependent manner (increasing risk from normoalbuminuria to severely increased albuminuria). APKD, Autosomal dominant polycystic kidney disease; BMI, body mass index; CMV, cytomegalovirus; DM, diabetes mellitus; HCV, hepatitis C; HLA, human leukocyte antigen; Hx, history; HypoMg, hypomagnesemia; IGT/IFG, impaired glucose tolerance/impaired fasting glucose; PTDM, posttransplantation diabetes mellitus.

NODAT back to PTDM, excluding transient posttransplantation hyperglycemia.²² Using the term NODAT is thought to be misleading because it seemingly excludes patients with pretransplant diabetes. Preexisting diabetes is often undiagnosed because of the effect of CKD on insulin metabolism and clearance and the lack of effective pretransplant screening.

Risk Factors

PTDM may arise from both transplant-related and traditional risk factors. The diabetogenic effect of corticosteroids, CNIs (tacrolimus > cyclosporine), and mTOR inhibitors have been well-described. Azathioprine, mycophenolic acid derivatives, and belatacept have not been shown to increase PTDM risk. Risk factors for PTDM are summarized in [Fig. 111.1](#). They can be loosely categorized into those that are nonmodifiable, potentially modifiable, and modifiable.

Management

Management of PTDM follows the conventional approach and clinical guidelines for diabetes in the general population. Medical management usually involves a multidisciplinary team approach involving patients (and frequently family members), transplant coordinators, transplant physicians, and diabetic educators. A shared decision-making approach by which clinicians and patients exchange information and reach a consensus on treatment decisions can be helpful.

The American Diabetes Association and European Association for the Study of Diabetes recommend a hemoglobin A1c (A1c) target of less than 7% in most patients to reduce the incidence of microvascular disease. More stringent control (A1c 6.0%–6.5%) might be considered in a subset of patients with short disease duration, long life expectancy,

and no significant CV disease, if this can be achieved without significant treatment-related adverse effects. For PTDM, the 2009 KDIGO clinical practice guidelines suggest a target A1c level from 7.0% to 7.5% but not less than 6.0%, particularly if hypoglycemic reactions are common.

Therapeutic Interventions

Nonpharmacologic Management

Obesity management is indicated because higher pretransplant BMI correlates with insulin resistance after transplantation. Lifestyle modifications in kidney transplant recipients include moderation of dietary sodium (<2400 mg sodium per day) and saturated fat intake (<7% calories from saturated fats, 2%–3% calories from trans-fatty acids), regular aerobic exercise, and weight reduction. Carbohydrate intake should be limited to 50% to 60% of caloric intake. The AHA guidelines for the general population recommend an intake of more than 25 g of dietary fiber per day and two servings of fish per week. Dietary guidelines for vegetarians are lacking. Flax seeds (or flaxseed oil) and chia seeds are alternative sources of omega-3 fatty acid. Dietitian referral is recommended. Defining realistic goals such as a target weight loss of 5% to 10% of total body weight and a patient-centered approach to education may be helpful.

Modification of Immunosuppression

Tacrolimus (TAC) has more consistently been shown to have a greater diabetogenic effect than CSA. A multicenter, prospective, randomized trial demonstrated that conversion from TAC to CSA resulted in reversal of diabetes, improved metabolic control, and reduced need for glucose-lowering therapy. At 12-month follow-up, 34% of patients

(14/41) in the CSA arm were free of diabetes compared with 10% (4/39) in the tacrolimus arm ($P = .01$). During the follow-up period, 39% of patients in the CSA arm discontinued glucose-lowering medication compared with 13% in the tacrolimus arm ($P = .01$). CSA conversion was associated with a mild increase in total and LDL-cholesterol and a greater reduction in eGFR (using the Modification of Diet in Renal Disease [MDRD] formula) compared with TAC maintenance immunosuppression at 12-month follow-up.²³ A meta-analysis of controlled clinical trials to assess the safety and efficacy of early steroid withdrawal or avoidance, steroid avoidance, or steroid withdrawal was associated with a decrease in PTDM incidence among CSA- but not TAC-treated kidney transplant recipients.²⁴ However, among CSA-treated patients, acute rejection episodes were more frequently observed in steroid avoidance compared with conventional steroid-treated groups.

A randomized trial of corticosteroid withdrawal (CSWD) demonstrated that CSWD has a limited impact in reducing PTDM compared with low-dose prednisone (5 mg/day from month 6–5 years). The proportion of PTDM patients requiring treatment was similar between CSWD and chronic corticosteroid (CCS) groups (CSWD 22.5% vs. CCS 21.5%). However, insulin use at 5 years was lower in the CSWD group (3.7% vs. 11.6%; $P = .049$).²⁵ In the current era of transplantation, most transplant centers advocate rapid steroid taper and lower target CSA and TAC levels, which may improve PTDM risk. However, modification of immunosuppression to alleviate the incidence of PTDM should be tailored to each patient, with the benefits of better glycemic control weighed against the risk for acute rejection. Furthermore, CSA use is associated with worse lipid profile compared with TAC therapy.

Pharmacologic Treatment

When lifestyle modification fails to achieve glycemic control, medical intervention is necessary. Metformin is not widely used in transplant recipients because of the concern for lactic acidosis in the presence of dynamic kidney allograft function, particularly in the early posttransplant period. However, the potential beneficial effects of metformin, including weight neutral or weight loss, cardioprotection, and lack of significant drug-drug interactions renders metformin an attractive treatment option for selected patients.²² Experimental studies suggest that sulfonylureas are associated with beta cell apoptosis and beta cell exhaustion, raising theoretical concern about their use in PTDM, particularly in the early posttransplantation period. In contrast, the dipeptidyl peptidase-4 (DPP-4) inhibitors preserve pancreatic beta cell function in diabetic animal models. DPP-4 inhibitors are also safe and effective in the treatment of PTDM. Studies in patients with type 2 DM (T2DM) demonstrated that GLP-1 receptor agonists have a beneficial effect on CVD and weight loss. However, delayed gastric emptying associated with their use raises the concern for erratic immunosuppressant drug absorption in the transplant setting. Studies evaluating the safety and efficacy of GLP-1 inhibitors in kidney transplant recipients are lacking. One small single-center retrospective study showed that GLP-1 receptor agonist use allowed insulin dose reduction and ameliorated hypoglycemia risk. Allograft function was stable and no impact on TAC dosing was observed during the 12-month study period ($n = 14$).²⁶ Further studies are needed. The sodium-glucose cotransporter type 2 (SGLT2) inhibitors are a new class of drugs that appear to reduce CV events and slow renal progression in the nontransplant population with diabetic CKD.²⁷ In a single-center, prospective, randomized, placebo-controlled trial involving 44 kidney transplant recipients with PTDM randomized to receive either empagliflozin ($n = 22$) or placebo ($n = 22$) for 24 weeks, both A1C and weight loss were found to be significantly reduced in empagliflozin-treated patients compared with their placebo-treated counterparts (-0.2% vs. 0.1% , $P = .025$ and -2.5 kg vs. an increase of 1.0 kg, $P = .014$, respectively).²⁸ No

significant differences in adverse events, immunosuppressive drug levels, or eGFR were observed between the two treatment groups except for one genital mycotic infection in the SGLT2 inhibitor treated group. Although the risk of genitourinary infectious complications associated with SGLT2 inhibitor use has not been consistently demonstrated, the US Food and Drug Administration warning labels regarding serious UTIs, including urosepsis and pyelonephritis, associated with its use remain in effect (updated March 2022). It is prudent to avoid SGLT2 inhibitor use in kidney transplant recipients with a history of recurrent UTIs, neurogenic bladder, or urologic abnormalities requiring chronic intermittent self-catheterization.

Evidence-based studies recommending one antidiabetic agent over the other in the context of transplantation are lacking. Hence, the choice should be based on the potential advantages and disadvantages of different classes of agents. Failure to achieve glycemic control despite glucose-lowering drug combination therapy requires initiation of basal-prandial insulin therapy. The 2014 international consensus guidelines on the screening, diagnosis, and management of early post-transplant hyperglycemia and PTDM is shown in Fig. 111.2.

Cigarette Smoking

Cigarette smoking in kidney transplant recipients is associated with perioperative complications, wound infections, increased risk of CVD, opportunistic infections, malignancies, and decreased patient and graft survival.²⁹ A multifaceted approach including behavioral and pharmacologic strategies appears to be most effective at getting patients to stop smoking.

Obesity

Single-center and large registry studies in kidney transplant recipients demonstrated that obesity (generally defined as body mass index [BMI] > 30 kg/m²) at transplant is associated with increased risk of wound and surgical site infections, delayed graft function (DGF), acute rejection episodes, and graft loss. Posttransplant obesity also increases the risk of heart failure, atrial fibrillation, CVD, and cardiac death after transplant. Management of posttransplantation obesity can be a challenge because weight gain is common after transplant because of improved appetite related to resolving uremia, use of corticosteroid, liberation of dietary restrictions, and an overall improved sense of well-being. Early initiation of healthy lifestyle changes and counseling by a dietitian are important for successful weight management. Corticosteroid reduction or withdrawal must be balanced against the risk of acute rejection and graft loss. The use of pharmacologic agents for weight reduction in the posttransplantation period is not recommended because of unknown potential drug-drug interactions. Data are inadequate regarding the safety and efficacy of posttransplantation gastric bypass surgery or adjustable gastric banding in ameliorating comorbid conditions such as hypertension, DM, and dyslipidemia. All studies reported to date were limited by lack of suitable comparison group, short follow-up, and heterogeneity in type of bariatric procedure and approach.³⁰ However, with refinement in surgical techniques and advancement in postoperative care, posttransplantation bariatric surgery for morbid obesity should be considered on a case-by-case basis. Laparoscopic procedures are generally associated with fewer complications compared with open procedures.

UNCONVENTIONAL CARDIOVASCULAR DISEASE RISK FACTORS

Proteinuria

Proteinuria occurs in 9% to 40% of kidney transplant recipients with a functioning graft. Proteinuria from native kidneys typically decreases

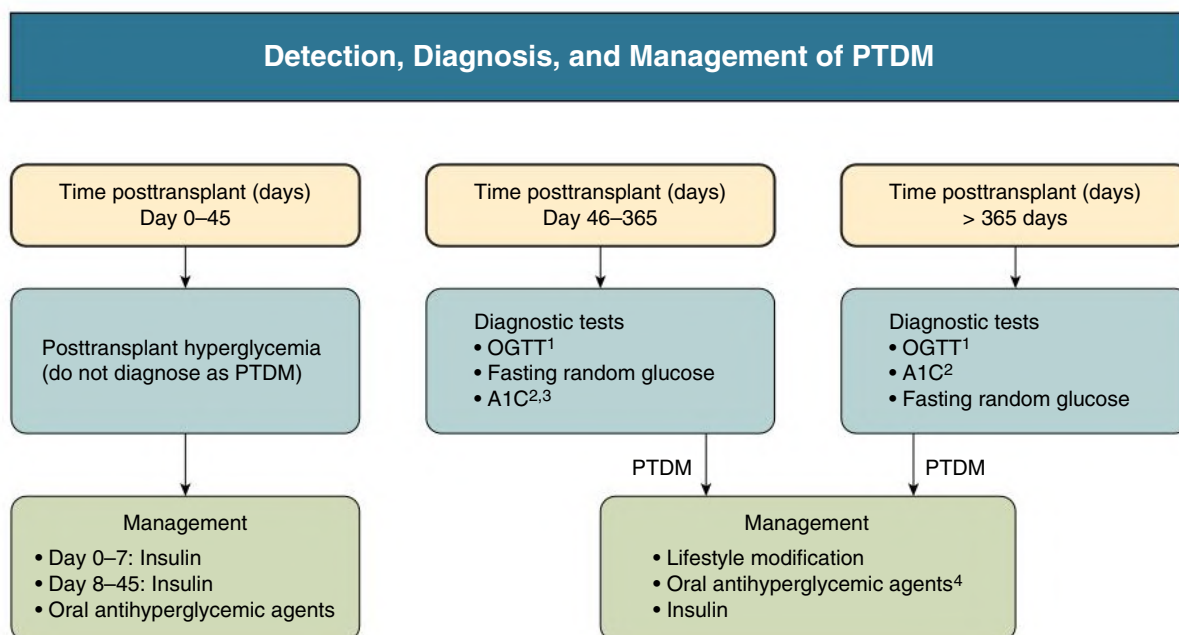


Fig. 111. 2 2014 International Consensus Guidelines on the Detection, Diagnosis, and Management of Posttransplantation Diabetes Mellitus (PTDM). ¹Seldom performed in clinical practice (time-consuming/cost). ²A1C cannot be accurately interpreted within the first 3 months after transplantation because anemia and impaired graft function can directly interfere with the A1C assay. Recent blood transfusion and dapsone may alter A1C level. ³A1C alone <365 days may underestimate PTDM and require confirmatory testing. ⁴Over the past several years, newer injectable antidiabetic agents have increasingly been used (however, it should also be noted that evidence-based recommendations are lacking). A1C, Hemoglobin A1C; OGTT, oral glucose tolerance test; PTDM, posttransplantation diabetes mellitus.

rapidly after transplantation and resolves within the first month after transplantation. Persistent or worsening proteinuria is usually indicative of allograft pathology. Hence, monitoring the urine protein-to-creatinine ratio at each clinic visit and allograft biopsy should be performed in kidney transplant recipients with persistent or worsening proteinuria and in those whose primary renal disease is focal segmental glomerulosclerosis, to monitor for disease recurrence. As in the nontransplant setting, posttransplantation proteinuria increases the risk for CV events, graft loss, and mortality. In a single-center cohort analysis of 137 heart transplant recipients, the development of proteinuria after CNI to sirolimus conversion therapy was found to be associated with diminished benefit on cardiac allograft vasculopathy progression and higher all-cause mortality risk compared with those without proteinuria (hazard ratio [HR], 3.8; $P = .01$).³¹ Whether mTOR inhibitor-induced proteinuria similarly increases mortality in kidney transplant recipients is unknown and warrants further studies. Although controlled trials evaluating the beneficial effect of treating proteinuria in reducing CV risk in kidney transplant recipients are lacking, ACE inhibitors or ARBs should be considered in transplant recipients with proteinuria because of their well-established antiproteinuric and cardioprotective effects.

Hyperhomocysteinemia

Although hyperhomocysteinemia has long been recognized as a risk factor for CAD, systematic review and meta-analysis showed no beneficial effect of homocysteine-lowering therapy on CVD risk reduction in patients with ESKD or in kidney transplant recipients.³² The use of homocysteine-lowering agents for the primary prevention of CVD is not recommended.⁸

Belatacept Use

Phase III RCTs comparing the safety and efficacy of belatacept-based versus cyclosporine-based immunosuppression in recipients of standard donor kidneys (BENEFIT study) and in those of extended-criteria donor

kidneys (BENEFIT-EXT study) demonstrated that, compared with cyclosporine, belatacept-based regimens resulted in (1) lower BP, cholesterol levels, and incidence of PTDM, (2) significantly reduced death and graft loss, and (3) improved long-term renal function. However, whether improvement in CV risk factor profile translates into reduction in CVD remains to be studied. In one single-center retrospective cohort study involving kidney transplant recipients at high CV risk, belatacept-based immunosuppression ($n = 35$) was not found to be associated with CV risk reduction compared with CNI-based therapy (control, $n = 150$). However, the study had several limitations, including retrospective nature, small sample size, different baseline characteristics, and duration of follow-up between the two treatment groups.³³

COMMON LABORATORY ABNORMALITIES

Anemia

In the immediate posttransplantation period, aggressive perioperative volume expansion may result in dilutional anemia. Refractory or severe anemia mandates aggressive evaluation to exclude the possibility of surgical postoperative bleeding, particularly in those with a rapid fall in hemoglobin (Hb) and hematocrit (Hct) levels.

Mild anemia is common in the early posttransplantation period when erythropoietin (EPO) therapy is typically discontinued but usually improves within several weeks to months. Suggested etiologic factors for posttransplantation anemia include iron deficiency, impaired graft function, acute rejection episodes, recent infection, and medications (e.g., azathioprine, mycophenolic acid derivatives, sirolimus, everolimus, dapsone, ACE inhibitors, and ARBs).

Assessment of baseline iron stores at the time of transplantation may be invaluable because iron deficiency is common in the dialysis population. Profound iron deficiency should be treated with intravenous (IV) iron as tolerated. The 2012 KDIGO anemia guidelines recommend initiating an erythropoiesis-stimulating agent in CKD patients

when Hb values are less than 9 to 10 g/dL, provided iron stores are adequate. A target Hb level of 10 to 11 g/dL is recommended. Large CKD anemia trials demonstrated a possible harmful effect of higher Hb levels and high epoetin doses. Observational studies in kidney transplant recipients similarly suggested that mortality may be increased with Hb levels greater than 12.5 g/dL. Although evidence-based recommendations are lacking, management of posttransplantation anemia to keep Hb level in the range of 10 to 11 g/dL seems reasonable.

Refractory anemia or anemia that fails to rise gradually to a normal or near-normal level after the first few posttransplantation weeks can be a result of occult gastrointestinal (GI) bleeding, tertiary hyperparathyroidism, underlying inflammatory conditions, or parvovirus B19 infection. EPO-resistant anemia has been described in patients receiving sirolimus immunosuppression. In stable kidney transplant recipients, conversion from sirolimus to mycophenolic acid derivatives may help resolve anemia. Although uncommon, drug-induced hemolysis from agents such as dapsone should also be considered.

Leukopenia and Thrombocytopenia

Leukopenia and thrombocytopenia are most commonly due to medications, including lymphocyte-depleting agents (e.g., thymoglobulin or alemtuzumab), azathioprine, mycophenolic acid derivatives, sirolimus, everolimus, and trimethoprim-sulfamethoxazole, among others. Withholding or reducing the dose of the offending agent generally corrects these abnormalities. Severe leukopenia may be safely treated with granulocyte-stimulating factor (G-CSF). Thrombotic microangiopathy (TMA) or cytomegalovirus (CMV) infection should be excluded. Parvovirus B19 infection may manifest with refractory anemia, pancytopenia, and TMA. Alemtuzumab can cause potentially fatal immune thrombocytopenia. Bortezomib has been reported to cause leukopenia and thrombocytopenia.

Posttransplantation Erythrocytosis

Posttransplantation erythrocytosis (PTE) may develop within the first 2 years and generally affects those with good allograft function. Its incidence appears to have decreased to less than 10% concomitant with the more frequent use of ACE inhibitors and ARBs. Spontaneous remission is observed in one-fourth of patients within 2 years from onset, whereas in the remaining three-fourths, erythrocytosis may persist for several years. Risk factors for PTE include the presence of native kidneys, male sex, absence of rejection episodes, high pretransplantation Hb, and hypertension. Smoking and diabetes have been shown to be risk factors for PTE in some but not all studies. Other suggested risk factors include polycystic kidney disease and glomerulonephritis as the cause of end-stage kidney disease (ESKD). Although transplant renal artery stenosis has not consistently been shown to be a risk factor for PTE, imaging studies to evaluate the iliac and renal arteries should be considered in patients with refractory PTE (see Chapter 43). In addition, the possibilities of renal cell carcinoma in the native or transplanted kidneys should be excluded.

Treatment is generally recommended for an Hb level greater than 17 to 18 g/dL or an Hct level greater than 51% to 54% because of the associated risk for thromboembolic complications, hypertension, and headaches. Hb and Hct treatment range may vary depending on sex and institutional reference range. Treatment with ACE inhibitors or ARBs is often sufficient, although phlebotomy may occasionally be necessary. Relapse is common and often necessitates long-term ACE inhibitor or ARB treatment. We often prescribe low-dose aspirin 81 mg daily unless contraindicated.

Hyperkalemia

Mild hyperkalemia is commonly encountered in kidney transplant recipients, particularly in the early posttransplantation period when

higher doses of CNI are used ($TAC > CSA$). It is often associated with mild hyperchloremic acidosis, a clinical presentation reminiscent of that of type 4 renal tubular acidosis. Other suggested mechanisms of CNI-induced hyperkalemia include activation of the sodium chloride cotransporter (NCC) in the distal convoluted tubule (leading to a decrease in the electrical gradient for potassium secretion via ROMK channels), inhibition of the ROMK channels and Na^+/K^+ -ATPase in the distal tubule, and/or downregulation of mineralocorticoid receptor expression.³⁴ In patients receiving CSA or TAC immunosuppression, a potassium level of 5.2 to 5.5 mmol/L is typical. Higher potassium levels, especially in the presence of concomitant use of drugs that may exacerbate hyperkalemia such as ACE inhibitors, ARBs, and β -blockers, may require their discontinuation. Caution is needed when potassium-containing phosphorus supplements are prescribed. Although both high-dose and standard-dose trimethoprim can cause hyperkalemia via an amiloride-like effect, the routine use of low-dose trimethoprim-sulfamethoxazole (TMP-SMX) prophylactic therapy is rarely the cause of severe or refractory hyperkalemia in kidney allograft recipients. Nonetheless, withholding of TMP-SMX should be considered in patients with severe or refractory hyperkalemia. In patients with recurrent hyperkalemia, consideration should be given to TMP-SMX dose reduction or TMP-SMX to dapsone or atovaquone conversion therapy (see Chapter 110 for *Pneumocystis jirovecii* prophylaxis treatment options). Treatment of hyperkalemia is discussed in Chapter 10. Sodium polystyrene sulfonate (Kayexalate) or calcium resonium enemas should be avoided in the early posttransplantation period to avoid colonic dilation and perforation. Although large-scale clinical trials in kidney transplant recipients are lacking, the newer cation exchangers patiromer and sodium zirconium cyclosilicate appear safe and effective in the treatment of hyperkalemia in kidney transplant recipients.³⁴ However, patiromer use in kidney transplant recipients may be hindered by potential drug-induced hypomagnesemia. This adverse drug effect may exacerbate the electrolyte abnormalities commonly seen in the posttransplantation period (discussed later). Limited studies suggest that patiromer may increase serum concentrations of TAC, whereas sodium zirconium cyclosilicate appears to have no impact on TAC pharmacokinetics.³⁴

Hypophosphatemia

Hypophosphatemia is frequently encountered in the first months after transplantation. Concomitant hypercalcemia suggests posttransplantation hyperparathyroidism. In the absence of hypercalcemia, renal phosphate-wasting syndrome or malnutrition should be considered. Early after transplantation, hypophosphatemia has been attributed to a massive initial diuresis, particularly after a living-donor kidney transplant, defective renal phosphate reabsorption caused by ischemic injury, glucosuria (caused by hyperglycemia-induced osmotic diuresis), and corticosteroid use, the latter by inhibiting proximal tubular reabsorption of phosphate. Fibroblast growth factor-23 (FGF-23) may contribute to early posttransplantation hypophosphatemia independent of parathyroid hormone (PTH) level, and pretransplant FGF-23 is the main predictor of posttransplant phosphate levels. In contrast to hypophosphatemia occurring in the early posttransplant period, persistent hypophosphatemia beyond 1 year after transplant is mainly because of persistent hyperparathyroidism rather than increased FGF-23 levels.

Hypercalcemia

Hypercalcemia is common after transplantation and is generally due to persistent secondary hyperparathyroidism. The concomitant presence of severe hypophosphatemia, particularly in patients

with excellent graft function, may exacerbate hypercalcemia through stimulation of renal proximal tubular 1 α -hydroxylase. Resolution of soft tissue calcifications and immobilization are potential contributing factors. Hypercalcemia may be masked by high-dose corticosteroid therapy but may manifest during tapering. In about two-thirds of patients, hypercalcemia resolves spontaneously within 6 to 12 months. However, spontaneous resolution occurs in less than half of those whose hypercalcemia existed before transplantation. Persistent hyperparathyroidism has generally been attributed to continued autonomous production of PTH from nodular hyperplastic glands, reduced density of calcitriol receptors, and decreased expression of the membrane calcium sensor receptors that render cells more resistant to physiologic concentrations of calcitriol and calcium. Persistently elevated FGF-23 levels after transplant may contribute to persistent hyperparathyroidism through inhibition of α 1-hydroxylase activity leading to low calcitriol levels. The risk for persistent hyperparathyroidism is increased with the duration of dialysis and the severity of pretransplant hyperparathyroidism. Severe and persistent hypercalcemia (>11.5 – 12 mg/dL [2.87 – 3.0 mmol/L]) for longer than 6 months in the first posttransplant year requires further evaluation. Initial assessment should include an intact PTH level. Parathyroid technetium-99m (^{99m}Tc)-sestamibi scan is generally required to determine whether the hyperparathyroidism arises from parathyroid adenoma, parathyroid gland hyperplasia, or hyperplastic nodular formation. Treatment with cinacalcet can be considered in kidney transplant recipients with hyperparathyroidism-associated hypercalcemia in the early posttransplant period. Serum calcium, phosphorus, and alkaline phosphatase should be monitored at each clinic visit. Because spontaneous resolution of hyperparathyroidism may take time, it is generally advisable to delay parathyroidectomy to 1 year after transplant. However, severe hypercalcemia associated with parathyroid adenoma often necessitates surgical parathyroidectomy. A multicenter, randomized, placebo-controlled trial demonstrated that cinacalcet use is safe and effective in the treatment of posttransplant hyperparathyroidism-associated hypercalcemia. Approximately 79% of cinacalcet, versus 3.5% of placebo-treated patients, achieved control of hypercalcemia (defined as calcium <10.2 mg/dL; $P < .001$). Markers of bone density (alkaline phosphatase, osteocalcin, urine N-telopeptide) and measured bone mineral density (BMD) at the femoral neck at 1 year were comparable between the two treatment groups.³⁵ Subtotal parathyroidectomy (or total parathyroidectomy with autotransplantation) is warranted in patients with tertiary hyperparathyroidism or persistent severe hypercalcemia refractory to medical management (>11.5 – 12 mg/dL [2.87 – 3.0 mmol/L]) for longer than 6 to 12 months, symptomatic or progressive hypercalcemia, nephrolithiasis, persistent osteitis fibrosa, calcium-related renal allograft dysfunction, or progressive vascular calcification and calciphylaxis.³⁶ Clinicians should remain vigilant for the potential development of nephrocalcinosis associated with the use of cinacalcet.

Hypomagnesemia

CSA, TAC, and sirolimus can cause hypomagnesemia by inducing urinary magnesium wasting. Other factors that may contribute to posttransplantation hypomagnesemia include loop diuretics, recovery from acute tubular necrosis, postobstructive polyuria, renal tubular acidosis, and proton pump inhibitors. Hypomagnesemia may be more common in diabetics. In the first 3 months after transplantation, a magnesium level of less than 1.5 mg/dL (0.62 mmol/L) is common.

Dietary magnesium intake is usually insufficient, and high-dose oral magnesium supplementation may be required. IV magnesium should be considered in patients with severe hypomagnesemia (<1.0 mg/dL or 0.41 mmol/L), particularly patients with a history of CAD or cardiac arrhythmias and those taking digoxin.

Abnormal Liver Function Tests

Elevation of liver enzymes is common in the early posttransplantation period and is generally caused by drug-related toxicity (Fig. 111.3). CSA and, less commonly, TAC may cause transient, self-limited, dose-dependent elevations of transaminase levels and mild hyperbilirubinemia secondary to defective bile secretion. Elevated liver enzymes caused by drug-related adverse effects (such as from statin use) generally improve or resolve after drug discontinuation or dose reduction.

Persistent or profound elevation in liver enzymes should prompt further evaluation to exclude infectious causes, including CMV, hepatitis B, and hepatitis C infections. Preemptive therapy to prevent posttransplantation reactivation of HBV and treatment of hepatitis B and C are discussed in Chapter 110. A suggested algorithm for the management of kidney transplant recipients with elevated liver enzymes is shown in Fig. 111.3.

BONE AND MINERAL METABOLISM AFTER KIDNEY TRANSPLANTATION

Posttransplantation bone disease is common after kidney transplantation because of corticosteroid use, hypophosphatemia, and disturbances in the FGF-23-PTH-vitamin D axis. Corticosteroids directly inhibit osteoblastogenesis, induce apoptosis of osteoblasts and osteocytes, and increase bone resorption through osteoclast activation. Other adverse effects of corticosteroids include inhibition of intestinal calcium absorption, enhancement of renal calcium excretion, and direct suppression of gonadal hormone secretion. Low plasma calcidiol (25(OH) vitamin D) level also may contribute to bone disease and is very common in kidney transplant recipients, with a reported prevalence of calcidiol deficiency of 30% (levels <20 ng/mL) and insufficiency of 81% (levels <30 ng/mL). Suggested causative factors include nutritional deficiency, malabsorption, decreased sun exposure, increased metabolism of calcidiol to calcitriol after a successful kidney transplant, and proteinuria (which may decrease levels of binding protein and total calcidiol levels). Other potentially causal factors include “normal” age-dependent osteoporosis, persistent metabolic acidosis, phosphate depletion, DM, hypogonadism, smoking, and CNI therapy. Experimental murine models demonstrated that CSA increases both bone formation and bone resorption, leading to a high-turnover bone loss, whereas TAC increases bone resorption without affecting bone formation.³⁷ In contrast, mTOR inhibitors may have a bone-sparing effect. Sirolimus was shown to inhibit osteoclast formation both in vitro and in vivo. One small European study showed that sirolimus decreased serum level of bone resorption markers in patients treated with sirolimus compared with those receiving a CNI-based regimen.³⁸

Osteoporosis

Posttransplantation decline in BMD is most pronounced in the first 6 months and correlates with higher corticosteroid exposure. This early rapid decrease in BMD is usually followed by a slower rate of bone loss, which reflects cumulative corticosteroid dose. Bone loss in the first posttransplantation year ranging from 4% to 10% at the lumbar spine

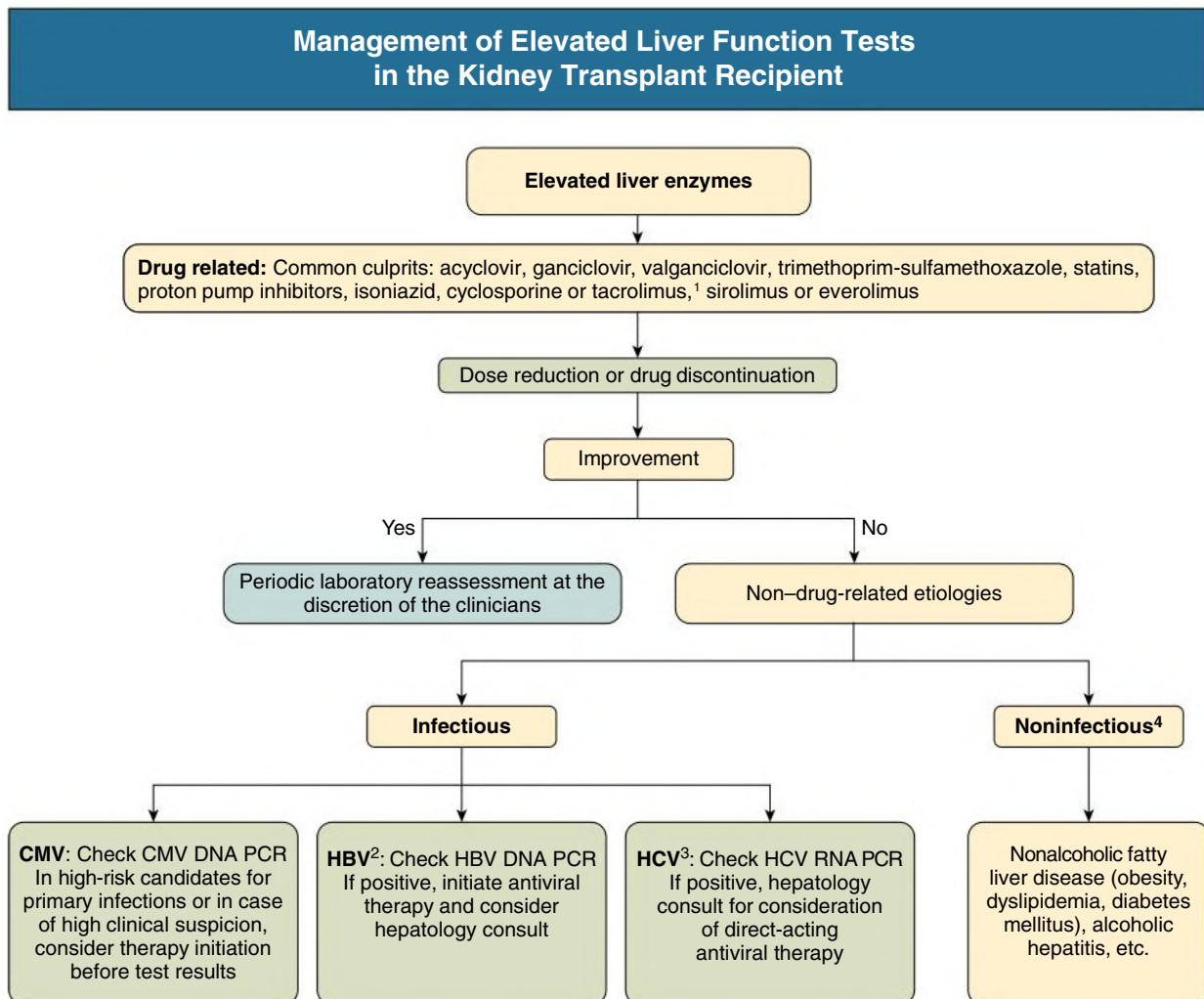


Fig. 111.3 Algorithm for the Management of Elevated Liver Enzymes in Kidney Transplant Recipients.

¹Cyclosporine and, less commonly, tacrolimus may cause transient, self-limited, dose-dependent elevation of aminotransferase levels and mild hyperbilirubinemia secondary to defective bile secretion. ²We advocate antiviral prophylactic therapy in all HBsAg-positive candidates at the time of transplantation. ³Treatment of HCV is discussed in Chapter 110. ⁴Appropriate evaluation and management similar to those in the nontransplant settings. CMV, Cytomegalovirus; HBV, hepatitis B; HCV, hepatitis C; PCR, polymerase chain reaction.

and 5% to 8% at the hip have been reported. However, in more recent years, more modest bone loss or even no loss have been noted, perhaps due to lower corticosteroid doses and increased use of calcium and vitamin D.³⁹ Evaluation of patients for bone loss or osteoporosis relies on dual-energy x-ray absorptiometry (DEXA) scan. One retrospective cohort study of 238 kidney transplant recipients with CKD stage 1 to 5 demonstrated that osteopenia (HR, 2.7; 95% CI, 1.6–4.6) and osteoporosis (HR, 3.5; 95% CI, 1.8–6.4) were associated with significantly increased hip fracture risk compared with normal BMD, independent of age, sex, and diabetes.⁴⁰ The mean eGFR at the time of DEXA testing was 48 mL/min/1.73 m² (range, 12–98 mL/min). In patients in the first 12 months after kidney transplant with eGFR greater than 30 mL/min/1.73 m² and low BMD, the 2017 KDOQI guideline suggests initiating treatment with vitamin D, calcitriol/alfacalcidol, and/or antiresorptive agents. The choice of treatment can be influenced by abnormal levels of calcium, phosphate, PTH, alkaline phosphatase, and 25-OH Vitamin D as well as bone biopsy if required. Data are insufficient to guide treatment after the first 12 months.

Avascular Necrosis

Historically, posttransplantation avascular necrosis (AVN) occurs in up to 15% to 20% and most commonly affects the femoral head and neck. More recently its prevalence has decreased to approximately 3% to 5% concomitant with the use of lower corticosteroid dose and rapid steroid tapering. AVN usually occurs within the first few years after transplantation and may affect other joints, including the knees, the shoulders, and, less commonly, the ankles, elbows, and wrists. Predisposing factors for AVN include greater exposure to IV corticosteroid pulse therapy, cumulative doses of more than 2 g for longer than 3 months, low bone mass, hyperparathyroidism, increasing dialysis duration, excessive weight gain, hyperlipidemia, microvascular thrombosis, and a history of local trauma. Early AVN of the femoral head commonly presents with hip or groin pain or referred knee pain. Magnetic resonance imaging (MRI) is the most sensitive technique for early detection, and plain radiographs are of limited value. Corticosteroid dose reduction or discontinuation has little if any effect on altering the course of established AVN and may jeopardize graft function.

Prevention and Management of Posttransplantation Bone Diseases

Management of posttransplantation bone disease has largely been based on studies involving postmenopausal osteoporosis and corticosteroid-induced osteopenia in nontransplant settings. Studies in solid organ transplantation have yielded variable and conflicting results. Nonetheless, preventive measures to minimize posttransplantation bone loss should be initiated early after transplantation. Early ambulation and regular weight-bearing exercise should be encouraged. Adequate calcium supplementation (1000 mg/day) in nonhypercalcemic patients is recommended in the first posttransplantation year to prevent rapid bone loss resulting from corticosteroid-induced decreased intestinal calcium absorption. The 2017 KDIGO guideline recommends measuring calcidiol levels and correcting vitamin D deficiency and insufficiency using treatment strategies recommended for the general population. In patients at increased risk for fracture, consideration should be given to rapid corticosteroid taper or corticosteroid-free immunosuppressive protocols after weighing the risks and benefits of acute rejection. High pretransplantation PTH levels predict persistent hyperparathyroidism after transplantation, which may contribute to bone loss after transplantation. Hence, preexisting hyperparathyroidism associated with CKD should be treated according to the KDIGO guideline.

In patients with preexisting secondary hyperparathyroidism, hypercalcemia and hypophosphatemia may develop after transplantation because of persistent hyperparathyroidism and the newly elevated calcitriol level. Conservative management with phosphate replacement and suppression of hyperparathyroidism with cinacalcet has generally been recommended. Although no guidelines exist, phosphate supplementation should be considered in patients with a phosphate level of up to 1.5 to 1.7 mg/dL (0.5 mmol/L) to avoid complications associated with severe hypophosphatemia (e.g., rhabdomyolysis, left ventricular dysfunction, respiratory muscle weakness, and hemolysis). The indication for treatment may depend on the institutional reference range for serum phosphate. Patients should be encouraged to increase phosphate-rich food intake, and phosphate replacement should be discontinued when serum phosphate stabilizes at around 2.0 mg/dL (0.64 mmol/L). Overaggressive phosphate replacement is not recommended because it can result in worsening existing hyperparathyroidism and persistently elevated FGF-23 levels. Phosphate supplementation may also increase the risk of vascular and kidney allograft calcifications.

Bisphosphonates increase BMD in postmenopausal females and patients with corticosteroid-induced osteoporosis, particularly at the lumbar spine and trochanter. Studies evaluating the safety and efficacy of bisphosphonates in the treatment and prevention of bone disease after kidney transplantation have yielded mixed and even contradictory results, in part because of differences in study design, concomitant use of calcium or vitamin D, timing of treatment, duration of follow-up, and preexisting renal osteodystrophy, among others. The benefit of bisphosphonates for BMD may be skeletal site-specific. Several meta-analyses of RCTs showed a significant improvement in BMD with bisphosphonates compared with controls in the lumbar spine and femoral neck. No effect on fracture risk was observed. However, the low incidence of fracture events, small sample sizes, and short follow-up duration in all these studies might limit the ability to draw conclusions on fracture risk.³⁹ The 2019 “Cochrane Kidney and Transplant Register of Studies” database involving RCTs and quasi-RCTs suggested that

bisphosphonate therapy may reduce fracture and bone pain after kidney transplantation among all age groups based on evidence of low certainty.⁴¹ Bisphosphonates should be used with caution in patients with impaired kidney function and avoided in CKD stages 4 to 5.

Denosumab is a humanized monoclonal antibody that binds to and prevents RANKL (receptor activator of nuclear factor kappa- β ligand) from activating its receptor (RANK) on the surface of osteoclasts and their precursors. Denosumab has been shown to increase BMD and decrease fracture risk in postmenopausal females. Limited studies in kidney transplant recipients suggest that denosumab in combination with calcium and vitamin D may improve BMD compared with calcium and vitamin D alone.

Estrogen or androgen deficiencies also have been suggested to contribute to posttransplantation bone loss. However, no data support or refute the use of hormonal therapy to prevent posttransplantation-related bone loss in postmenopausal females or in males with hypogonadism; hence, treatment should be individualized.

Teriparatide, a recombinant human parathyroid hormone 1 to 34, is the only osteoanabolic agent approved by the FDA for use in corticosteroid-induced osteoporosis in the general population. Little evidence exists to support its use after kidney transplantation.⁴²

In summary, calcium supplementation, vitamin D, and bisphosphonates remain the backbone of treatment of abnormal bone and mineral metabolism after kidney transplantation. Persistent hyperparathyroidism or hypophosphatemia should be treated before considering additional therapies for osteoporosis. In the absence of evidenced-based recommendations, the use of bisphosphonates in the posttransplantation setting should be individualized. In our opinion, bisphosphonate therapy may be justifiable in high-risk individuals, including those with preexisting osteoporosis or documented overall low bone mass, those who are receiving high-dose corticosteroids or those with history of fractures, and those who are postmenopausal. Biochemical markers of low bone turnover (serum PTH and bone-specific alkaline phosphatase levels below or at the lower limits of normal) may be used to guide therapy. In high-risk individuals who are intolerant of bisphosphonates, denosumab can be used as an alternative treatment option. Profound hypocalcemia associated with the use of denosumab, particularly in patients with low GFR, has been reported and should not be overlooked.

GOUT

Potential risk factors for the development of posttransplantation hyperuricemia and gouty arthritis include pretransplantation hyperuricemia, graft impairment, obesity, diuretic use, and CSA compared with TAC. CSA may impair renal excretion of uric acid secondary to decreased GFR and increase net uric acid reabsorption by the proximal tubule.

Management of the acute gouty attack includes topical ice and rest of the inflamed joint. Pharmacologic treatments include colchicine, increased corticosteroid dose, or nonsteroidal anti-inflammatory drugs (NSAIDs). However, NSAIDs should be avoided unless absolutely necessary. Progressive steroid tapering to low-dose maintenance antirejection therapy (usually over 2 to 3 weeks) is recommended to prevent rebound gout flare.

Management of chronic gout is directed at lowering uric acid levels. Allopurinol, a xanthine oxidase inhibitor, should be started at 100 to 200 mg/day, particularly in the presence of impaired

allograft function, because lower eGFR predisposes to severe allopurinol toxicity from retention of the metabolite oxypurinol. The dose can be increased by 50-mg increment (adjusted for renal function) every 4 weeks to maintain uric acid level less than 6 mg/dL in symptomatic patients. We do not recommend allopurinol in asymptomatic patients based on uric acid level alone. Febuxostat, a nonpurine analog xanthine oxidase inhibitor, can be considered in allopurinol-allergic patients. It is administered as 40 or 80 mg/day with no dose adjustment required for patients with renal impairment. However, one large randomized trial involving high CV risk patients (defined as patients with a history of major cardiovascular disease before randomization) demonstrated higher all-cause and CV mortality among febuxostat-treated compared with allopurinol-treated patients.⁴³ Both allopurinol and febuxostat should be used cautiously in patients taking azathioprine because of the inhibition of azathioprine metabolism by xanthine oxidase inhibitors (e.g., when used in combination therapy with allopurinol or febuxostat, consider azathioprine dose reduction by 25% and close monitoring of complete blood counts). Alternatively, azathioprine to mycophenolic acid derivative conversion therapy may be considered. The latter does not interact with allopurinol or febuxostat. Recombinant pegylated uricase (pegloticase), which converts uric acid to the more soluble and readily excreted purine end-product allantoin, rapidly reduces uric acid levels in patients refractory to conventional therapy. Pegloticase was approved by the FDA in 2010 for treatment of refractory gout, but high titers of antibodies against the drug is associated with loss of efficacy and

infusion reactions. Recent studies suggest that posttransplantation immunosuppressive therapy may reduce immunologic response to pegloticase therapy by inhibiting the development of antipegloticase antibodies. Successful treatment of a heart transplant recipient with polyarticular tophaceous gout and hyperuricemia with a 38-week course of pegloticase has been reported.⁴⁴ Whether immunosuppressive therapy in the transplant setting may prevent antipegloticase antibody development and retain the urate-lowering efficacy among organ transplant recipients warrants further studies.

OUTPATIENT CARE

We recommend that patients be seen two times a week for the first 4 weeks after transplantation and weekly for the next month. After the first 2 months, the frequency of outpatient visits depends on the complexity of the patients' early postoperative course. Patients with stable graft function and an uneventful postoperative course can return to work or their regular daily activities 2 to 3 months after transplantation. Laboratory assessment during the first month after transplantation should include serum creatinine and electrolyte values, fasting glucose level, liver enzymes, immunosuppressive drug levels, and complete blood count with platelets. Urinalysis and, if clinically indicated, urine culture, and urine protein-to-creatinine ratio (see proteinuria section) also should be performed. At our institution, the patient's care will generally be returned to the referring nephrologist after 3 months. However, continued annual follow-up at the transplantation center is recommended.

SELF-ASSESSMENT QUESTIONS

1. Management of posttransplantation hypertension should include lifestyle modifications and which one of the following?
 - A. Stringent blood pressure control to a goal of less than 130/80 mm HG should begin in the early posttransplantation period.
 - B. An ACE inhibitor is the antihypertensive agent of choice after kidney transplantation because it improves patient and graft survival.
 - C. The KDIGO guidelines suggest a blood pressure goal of less than 130/80 mm Hg if albuminuria is present.
 - D. None of the above.
2. Which of the following drug combination therapies is more likely to cause myopathy and rhabdomyolysis?
 - A. High-dose statin and ezetimibe
 - B. High-dose statin and tacrolimus
 - C. High-dose statin and cyclosporine
 - D. Both B and C
3. Which one of the following statements is *correct*?
 - A. Cyclosporine (CSA) is more diabetogenic than tacrolimus (TAC).
 - B. Belatacept is less diabetogenic than both TAC and CSA.
 - C. Unlike the calcineurin inhibitors (CSA and TAC), mTOR inhibitors (sirolimus and everolimus) are nondiabetogenic.
 - D. Among currently available immunosuppressive agents, mTOR inhibitor use is associated with the worst lipid profile followed by CSA and TAC.
 - E. The mycophenolic acid derivatives mycophenolate mofetil and mycophenolate sodium are less diabetogenic than azathioprine.

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